KG COLLEGE OF ARTS AND SCIENCE
Autonomous Institution | Affiliated to Bharathiar University
Accredited with A++ Grade by NAAC
ISO 9001:2015 Certified Institution
KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

Event Registration and Participant Management System

Submitted by

NEVETHAA R S

25BCS036

Under the Guidance of

Dr . BOOPALAN SUNDARAMOORTHY

Technical Head

Technical Department, KG College of Arts and Science

Academic Year: **2025 – 2026**

APRIL 2026

Semester: **II**

DECLARATION

I hereby declare that the project entitled "**EVENT REGISTRATION AND PARTICIPANT MANAGEMENT SYSTEM"** submitted in partial fulfillment of the requirements for the course **Java programming language** is a **record of my original work** carried out under the guidance of **Dr.Boopalan Sundaramoorthy**, Technical Head, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

_____

NEVETHAA R S

CERTIFICATE

This is to certify that the project entitled **"Event Registrtion and Participant Management System"** is a **bona fide work** carried out by **Nevethaa.R.S, I B.Sc CS [2025 - 2028]**, **Department Of Computer Science**, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **Java Programming Language**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: COIMABATORE

Date: 10/04/2026

Submitted for the Viva-Voce Examination held on **10/04/2026**

Guide:
(Technical Head, KGCAS)

_____

Head of the Department
(Technical Head, KGCAS)

_____

TABLE OF CONTENTS

S. No.	Topics
1	**Introduction**
1.1	Objective
1.2	Overview
1.3	Features
2	**Program Analysis**
2.1	Problem Definition
2.2	Project Overview
2.3	Scopes and Limitations
3	**System Design**
3.1	Modules
4	**Methodology**
4.1	Problem Analysis
4.2	System Design
4.3	Program Development
4.4	Testing and Validation
4.5	Result and Documentation
4.6	Algorithm
4.7	Flowchart
5	**Source Code**
6	**Input and Output Screens**
7	**Discussion and Conclusion**
7.1	Discussion
7.2	Conclusion
8	**References**

ABSTRACT

The **Event Registration and Participant Management System** is a Java-based application developed using Java Swing to provide a user-friendly graphical interface for managing event registrations. In many colleges and organizations, participant registration is handled manually, which can result in data duplication, errors, and difficulty in searching and managing records. This project aims to automate and simplify the entire registration and management process.

The system allows users to register participants by entering details such as participant ID, name, college, event name, and contact number. It includes features like adding new participants, updating existing records, deleting entries, and searching participants using partial name or ID matching. The application also provides event-based filtering and a dashboard that displays the total number of participants and event-wise participation count.

The project is implemented using **Core Java programming concepts** such as classes, objects, ArrayList collections, file handling (serialization), exception handling, and event-driven programming with **Java Swing components** like JTable, JTextField, and JButton. The system ensures data persistence by storing participant details in a file, allowing data to be saved and loaded efficiently.

The main objective of this system is to improve accuracy in data management, reduce manual effort, and provide quick and efficient access to participant information. It enhances user experience through features like keyboard navigation, automatic text formatting, and highlighted search results.

This project demonstrates the practical application of **object-oriented programming concepts and GUI development in Java** and serves as a foundation for building more advanced event management systems with additional features like online registration, database integration, and real-time analytics.

1. INTRODUCTION

In educational institutions and organizations, managing event registrations efficiently is essential for conducting successful events and ensuring smooth coordination. Traditionally, many colleges handle participant registration manually using paper forms or basic spreadsheets. This manual process is time-consuming and often leads to issues such as data duplication, errors in entry, difficulty in searching records, and poor data management.

With the advancement of computer technology, automated registration systems have become necessary to improve efficiency and accuracy in handling participant data. A computerized system allows event organizers to easily register participants, maintain records systematically, and quickly access required information without confusion or delay.

The **Event Registration and Participant Management System** is a Java-based application designed to automate the process of participant registration and data management. The system enables users to enter participant details such as ID, name, college, event, and contact number. It provides functionalities to add, update, delete, search, and filter participant records efficiently. The application also includes features like partial search, event-wise filtering, and a dashboard that displays participant statistics.

This project is developed using **Core Java programming concepts** such as classes, objects, ArrayList collections, file handling (serialization), exception handling, and event-driven programming using Java Swing for building a **graphical user interface (GUI)**. The system ensures data persistence by saving participant details into a file, allowing easy retrieval and management of records.

The primary goal of this system is to reduce manual workload, improve accuracy in data handling, and provide a faster and more efficient way to manage event registrations. This project also demonstrates how Java can be used to develop practical real-world applications and serves as a foundation for building more advanced event management systems in the future.

1.1 Objective

The main objective of the Event Registration and Participant Management System is to develop a simple and efficient application using **Core Java** that automates the process of **participant registration** and **data management** for events.

The specific objectives of this project are:

- To design and develop a **Java-based system**
- To automate **participant data management**
- To reduce **manual errors** in data entry
- To provide a **user-friendly interface** using **Java Swing**
- To enable **efficient searching** using **partial name or ID**
- To allow **update and delete operations**
- To implement **event-wise filtering**
- To display **participant statistics** using a **dashboard**
- To ensure **data persistence** using **file handling (serialization)**
- To demonstrate **Core Java concepts** such as **classes, objects, ArrayList, loops, and exception handling.**

1.2 Overview

The **Event Registration and Participant Management System** is a **menu-driven application** developed using the **Java programming language**. The system provides a **digital platform** where **event organizers** can **register participants** and manage their details efficiently.

The system allows users to enter **participant information** such as **ID, name, college, event, and contact number**. It provides options to **add**, **update**, **delete**, and **search** participant records using **name or ID**. The application also supports **partial search functionality**, making it easier to find records quickly.

The system displays all participant details in a **structured table format** and includes features like **event-wise filtering** and a **dashboard** to view **participant statistics**. All data is stored using **file handling techniques**, ensuring that records are **saved and retrieved** whenever required.

The system helps simplify the registration process by eliminating **manual record-keeping** and reducing the chances of **errors**. The application is developed using **Java Swing** to provide a **graphical user interface (GUI)** that is easy to use and interactive.

This project demonstrates the practical use of **Java programming** in developing **real-world applications** for managing events in **colleges and organizations**.

1.3 Features

The **Event Registration and Participant Management System** includes the following main features:

• **Participant Registration** – Allows users to enter and store **participant details** such as **ID, name, college, event, and contact number**.

• **Add / Update Functionality** – Enables users to **add new records** and **update existing participant details** easily.

• **Delete Option** – Provides the ability to **remove participant records** using **ID or name**.

• **Search Functionality** – Allows users to **search participants** using **ID or name**, including **partial search** support.

• **Table Display** – Displays all participant records in a **structured table format (JTable)** for easy viewing.

• **Event-wise Filtering** – Enables users to **view participants based on a specific event**.

• **Dashboard View** – Displays **participant statistics** such as **total participants** and **event-wise count**.

• **Data Persistence** – Stores data using **file handling (serialization)** to ensure records are **saved and retrieved**.

• **User-Friendly Interface** – Provides an **interactive GUI** using **Java Swing components**.

• **Keyboard Navigation** – Supports **easy navigation using keyboard keys** for faster data entry.

• **Input Validation** – Ensures **correct data entry** (e.g., numeric ID, valid contact number).

• **Fast and Efficient Management** – Reduces **manual effort** and improves **data handling efficiency**.

2. PROGRAM ANALYSIS

Program Analysis is an important phase in software development where the problem and system requirements are identified. It helps in understanding how the Event Registration System should function effectively.

This phase identifies issues in the manual system such as data duplication, record errors, and difficulty in searching. It proposes a computerized solution to improve accuracy and efficiency.

The system supports operations like registration, searching, updating, deleting, and event filtering, ensuring quick data access and reduced manual work.

2.1 Problem Definition

In many colleges and organizations, event registration and participant management are often handled manually using paper forms or basic spreadsheets. This traditional method creates several challenges.

Some common problems include:
- Errors in data entry
- Time-consuming registration process
- Difficulty in searching participant records
- Duplication of data
- Poor management of event-wise participant details

Manual systems also increase the workload for event organizers and can lead to confusion and inefficiency, especially when handling a large number of participants.

Therefore, there is a need for an automated system that can efficiently manage participant registration, store data properly, and provide quick access to information.

The **Event Registration and Participant Management System** is designed to solve these problems by providing a computerized solution that automates the registration and data management process.

2.2 Project Overview

The **Event Registration and Participant Management System** is a Java-based application developed to simplify the process of managing event registrations in colleges and organizations.

The system provides a user-friendly interface where users can enter participant details such as ID, name, college, event, and contact number. The system automatically stores and manages the data, allowing easy access and modification of participant records.

The system includes the following functionalities:
- Registering participants with required details
- Displaying all participant records in a table format
- Searching participants using ID or name
- Updating existing participant information
- Deleting participant records
- Filtering participants based on event
- Viewing participant statistics through a dashboard

This project is implemented using **Core Java programming concepts** and is developed using **Java Swing to provide a graphical user interface** for easy and efficient interaction.

2.3 Scopes and Limitations

Scope

The scope of the **Event Registration and Participant Management System** includes the following:
- Managing participant registration details
- Storing and organizing participant data efficiently
- Searching and retrieving participant information
- Updating and deleting participant records
- Filtering participants based on events
- Displaying participant statistics through a dashboard
- Demonstrating the practical use of Java programming concepts

The system can be used in:
- Colleges
- Universities
- Event management teams
- Workshops and seminars
- Cultural and technical events

Limitations:

- The system uses file handling instead of a database, which is not suitable for very large data
- No online registration or web-based access
- Limited to single-user operation
- No advanced security features
- Requires manual data entry

3. SYSTEM DESIGN

System design is the phase where the **structure** and **workflow** of the system are planned before implementation. It helps in organizing the program efficiently.

The **Event Registration and Participant Management System** is designed as a **user-friendly application** to handle participant registration and data management.

The system is divided into **modules** such as registration, search, update, delete, filtering, and dashboard, where each module performs a specific function.

3.1 Modules

The main modules of the Event Registration and Participant Management System are explained below:

1. Participant Registration Module
This module is responsible for entering and storing participant details such as ID, name, college, event, and contact number. It helps users register new participants into the system.

Functions of this module include:
- Accepting participant details
- Validating input (ID must be numeric, contact must be 10 digits)
- Storing participant data in the system

The data is saved so that it can be used later for management and retrieval.

---

2. Display Module
This module displays all registered participant records in a table format.

Functions of this module include:
- Showing all participant details
- Organizing data in rows and columns
- Providing a clear view of stored records

The table is displayed when the system runs and updates automatically after every operation.

---

3. Search Module
This module allows users to search for participants using ID or name.

Functions of this module include:
• Accepting search input
• Supporting partial name matching
• Highlighting matching records

The system helps users quickly find participant details without scanning the entire list.

---

4. Update Module
This module is used to modify existing participant information.

Functions of this module include:
• Selecting a record from the table
• Editing participant details
• Saving updated information

The system ensures that updated data replaces old records correctly.

---

5. Delete Module
This module allows users to remove participant records from the system.

Functions of this module include:
• Deleting records using ID or name
• Removing selected participant details
• Updating the system after deletion

This helps in maintaining accurate and clean data.

---

6. Event-wise Filter Module
This module displays participants based on a specific event.

Functions of this module include:
• Accepting event name as input
• Filtering participant records
• Displaying only matching results

This helps organizers analyze participation in each event.

---

7. Dashboard Module

This module provides an overview of participant statistics.

Functions of this module include:

- Displaying total number of participants
- Showing event-wise participant count
- Providing quick summary information

The dashboard helps in better decision-making and analysis.

---

8. Exit Module

This module allows the user to safely close the application.

When the user exits the system, the program terminates after saving the data.

4. METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system.

For the **Event Registration and Participant Management System**, a structured development approach was followed to ensure that the system meets the requirements and functions efficiently.

The development process includes the following stages:

- Problem Analysis
- System Design
- Program Development
- Testing and Validation
- Result and Documentation

4.1. Problem Analysis

In this phase, the problem is studied and analyzed in detail. The requirements of the system are identified based on the needs of event registration and participant management.

The main problems identified in manual systems are:

- Errors in data entry may lead to incorrect records
- Registration process is time-consuming
- Difficult to search and manage participant details
- Data duplication and poor record maintenance

Based on these problems, the system requirements were defined to develop an automated application that simplifies participant registration and improves data management efficiency.

4.2. System Design

In this stage, the overall structure of the system is designed before coding begins.

The system is divided into modules such as:

- Participant Registration Module
- Display Module
- Search Module
- Update Module
- Delete Module
- Event-wise Filter Module
- Dashboard Module
- Exit Module

The program flow and structure are designed so that users can easily interact with the system.

4.3. Program Development

The Event Registration and Participant Management System is a Java-based application developed to simplify the process of managing event registrations in colleges and organizations.

The system provides a user-friendly interface where users can enter participant details such as ID, name, college, event, and contact number. The system automatically stores and manages the data, allowing easy access and modification of participant records.

The system includes the following functionalities:

- Registering participants with required details
- Displaying all participant records in a table format
- Searching participants using ID or name
- Updating existing participant information
- Deleting participant records
- Filtering participants based on event
- Viewing participant statistics through a dashboard

This project is implemented using Core Java programming concepts and is developed using Java Swing to provide a graphical user interface for easy and efficient interaction.

4.4. Testing and Validation

Testing ensures that the program works correctly and produces the expected results.

Different test cases were used to check the functionality of the system.

Test Case	**Expected Result**
Register participant	Participant details saved correctly
Search participant by ID	Correct participant displayed
Update participant details	Changes saved and displayed correctly
Delete participant	Record removed from the system
Filter by event	Only participants of the selected event displayed
Invalid input (ID/contact)	Error handled properly

This phase ensures that the program runs without errors and produces accurate results.

4.5. Result and Documentation

After successful testing, the results of the system are documented. The system manages participant registration, updates, deletion, and displays records accurately.

The results are demonstrated through **program execution** and **screenshots** showing the input and output of the system, such as registered participants, search results, and dashboard statistics.

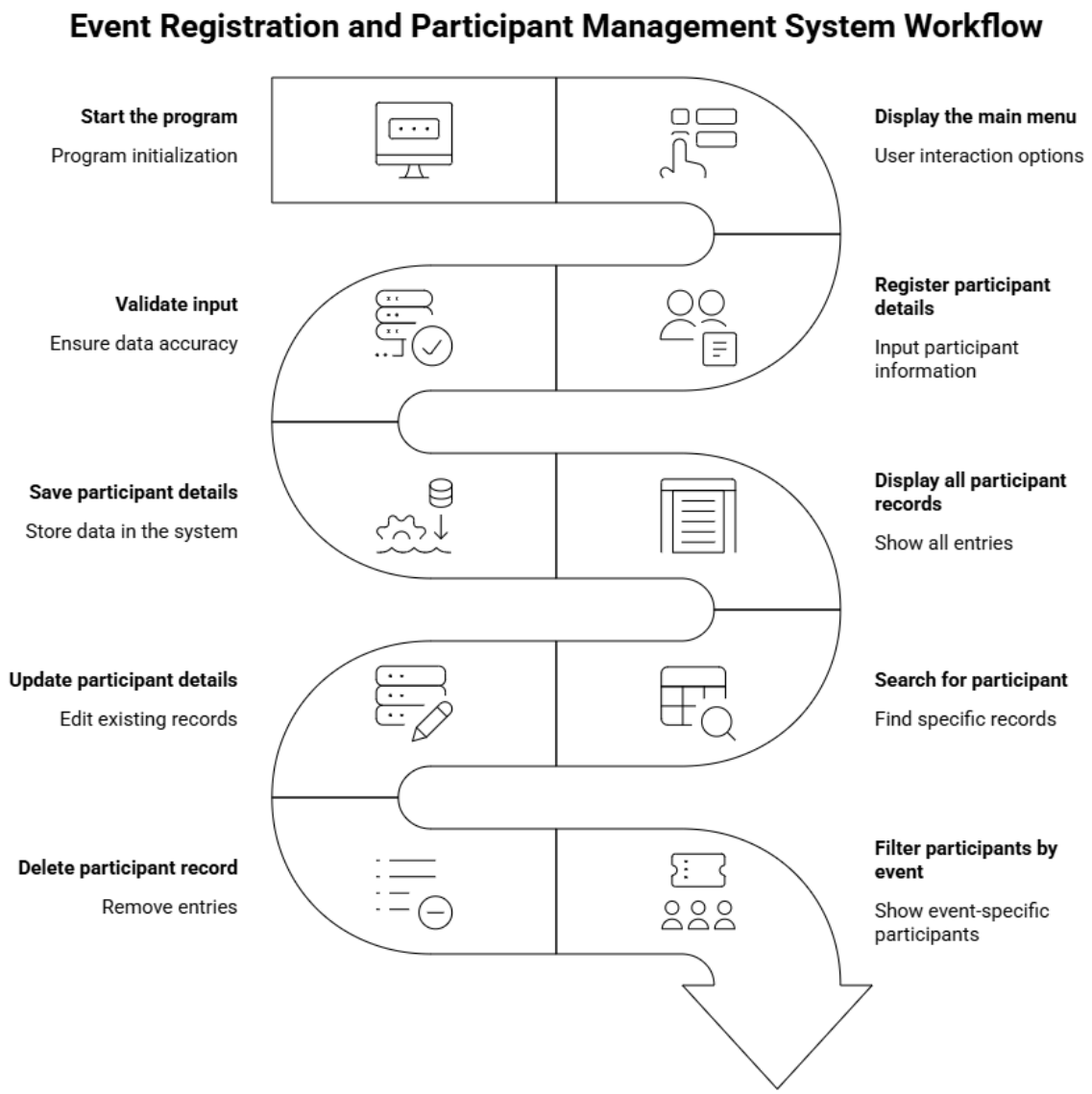
Proper documentation is prepared to explain the **design**, **implementation**, and **working** of the system, providing a complete understanding of how the Event Registration and Participant Management System functions.

4.6. Algorithm

Algorithm for the Event Registration and Participant Management System:

1. Start the program.
2. Display the main menu with options: Register, Display, Search, Update, Delete, Filter, Dashboard, Exit.
3. If the user selects **Register**, accept participant details: ID, Name, College, Event, Contact.
4. Validate the input (ID must be numeric, Contact must be 10 digits).
5. Save the participant details in the system.
6. If the user selects **Display**, show all participant records in a table.
7. If the user selects **Search**, accept ID or Name and display matching participant(s).
8. If the user selects **Update**, allow editing of selected participant details and save changes.
9. If the user selects **Delete**, remove participant record by ID or Name.
10. If the user selects **Filter**, accept an event name and display participants of that event only.
11. If the user selects **Dashboard**, display total participants and event-wise counts.
12. If the user selects **Exit**, save all data and terminate the program.
13. Repeat steps 2–12 until the user chooses to exit.
14. End the program.

4.7. Flowchart



5. SOURCE CODE

1. Main Class

This is where the program starts and initializes the GUI.

```
public class EventGUI extends JFrame {
   public static void main(String[] args) {
      // Launch the Event Registration System
      new EventGUI();
   }
}
```

Explanation:

- `EventGUI` class contains the GUI and all system functionalities.
- `main` method creates an instance of the system to start the program.

2. Menu / User Interface Logic

This part handles the **menu-driven interface** for user interaction.

```
        // Creating buttons for system menu
        JButton add = new JButton("Register");
        JButton clear = new JButton("Clear");
        JButton search = new JButton("Search");
        JButton delete = new JButton("Delete");
        JButton filter = new JButton("View by Event");
        JButton dashboard = new JButton("Dashboard");
        JButton showAll = new JButton("Show All");
        // Adding action listeners to handle user selection
        add.addActionListener(e -> addParticipant());
        clear.addActionListener(e -> clearFields());
        search.addActionListener(e -> search());
```

```
delete.addActionListener(e -> delete());

filter.addActionListener(e -> viewByEvent());

dashboard.addActionListener(e -> showDashboard());

showAll.addActionListener(e -> refresh());
```

Explanation:

- Buttons act as the **menu options** for the user.
- Each button calls a **specific method** to perform operations like add, delete, search, filter, or display dashboard.

3. Participant Management Logic

This part handles **adding, updating, deleting, and managing participant records**.

```
// Adding participant

void addParticipant() {

  try {

    int id = Integer.parseInt(txtId.getText());

    String name = txtName.getText();

    String college = txtCollege.getText();

    String event = txtEvent.getText();

    String contact = txtContact.getText();

    // Input validation

    if (!contact.matches("\\d{10}")) {

      JOptionPane.showMessageDialog(this, "Contact must be 10 digits!");

      return;

    }

    // Add participant to list

    list.add(new Participant(id, name, college, event, contact));

    save(); // Save to file
```

```
            refresh(); // Update table
            clearFields(); // Clear input fields
            JOptionPane.showMessageDialog(this, "Registered Successfully!");
        } catch (Exception e) {
            JOptionPane.showMessageDialog(this, "Error!");
        }
    }
    // Display participants in table
    void refresh() {
        model.setRowCount(0);
        for (Participant p : list) {
            model.addRow(new Object[]{p.id, p.name, p.college, p.event, p.contact});
        }
    }
    // Search participant by ID or Name
    void search() {
        String input = txtSearch.getText().trim().toLowerCase();
        table.clearSelection();
        for (int i = 0; i < model.getRowCount(); i++) {
            String id = model.getValueAt(i, 0).toString();
            String name = model.getValueAt(i, 1).toString().toLowerCase();
            if (id.contains(input) || name.contains(input)) {
                table.addRowSelectionInterval(i, i);
                table.scrollRectToVisible(table.getCellRect(i, 0, true));
            }
```

```
    }

}
```

Explanation:

- `addParticipant()` handles **adding new participants** with validation.
- `refresh()` updates the **JTable** to show current participants.
- `search()` allows **searching participants** by ID or name.
- Other similar methods handle **update, delete, filtering, and dashboard statistics**.

6. INPUT AND OUTPUT SCREENS

Figure 1: Main Menu Screen

Shows the menu options for registration, search, update, delete, filter, dashboard, and exit.

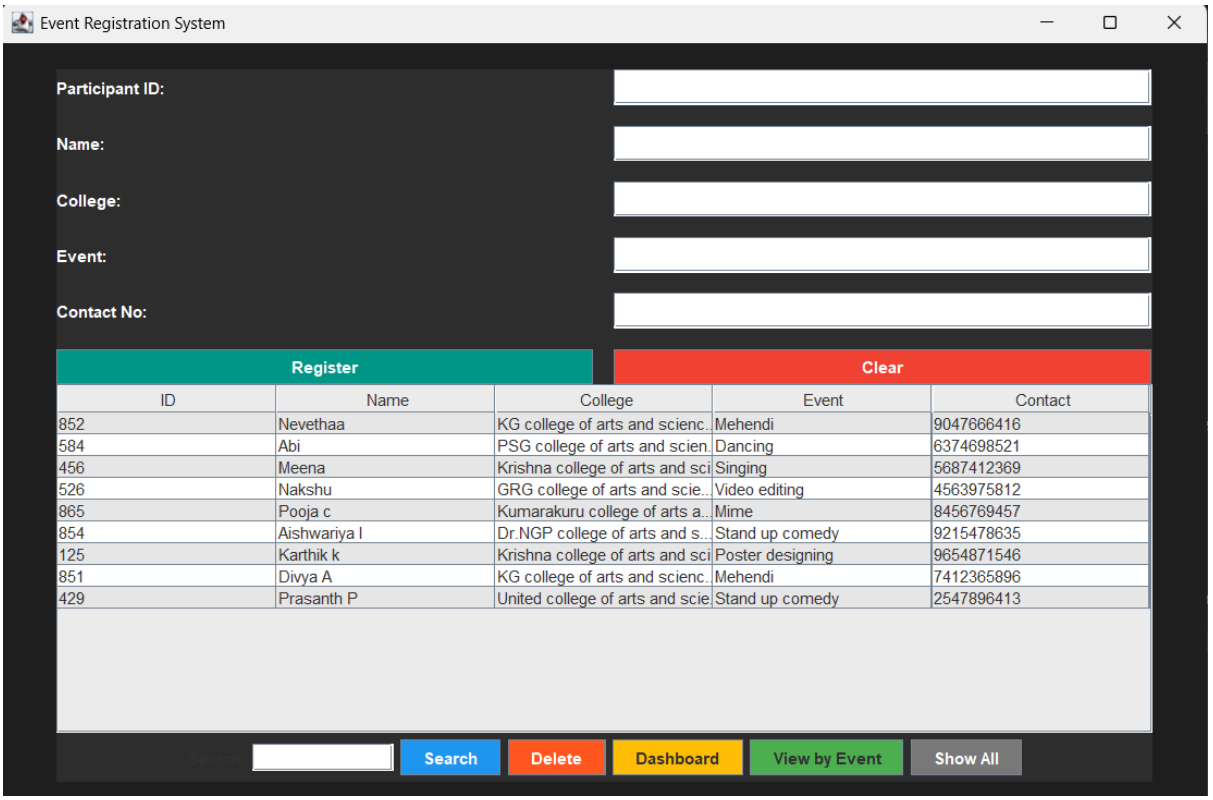


Figure 2: Participant Registration Screen

Shows the input fields for entering participant details: ID, Name, College, Event, Contact.

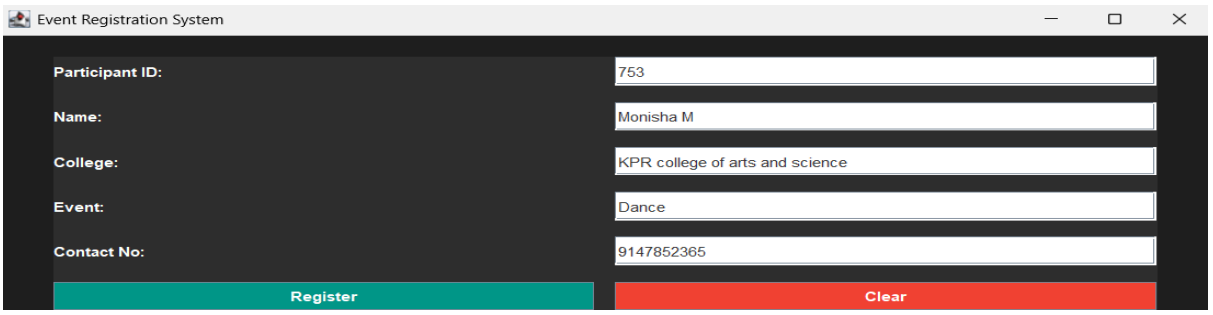


Figure 3: Display / Table Output Screen

Shows the registered participants in a table format after registration.

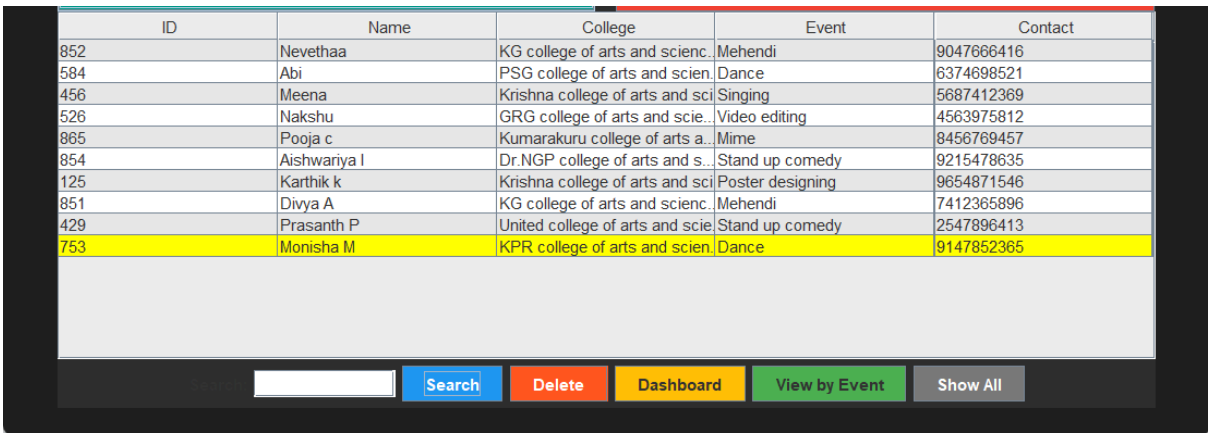
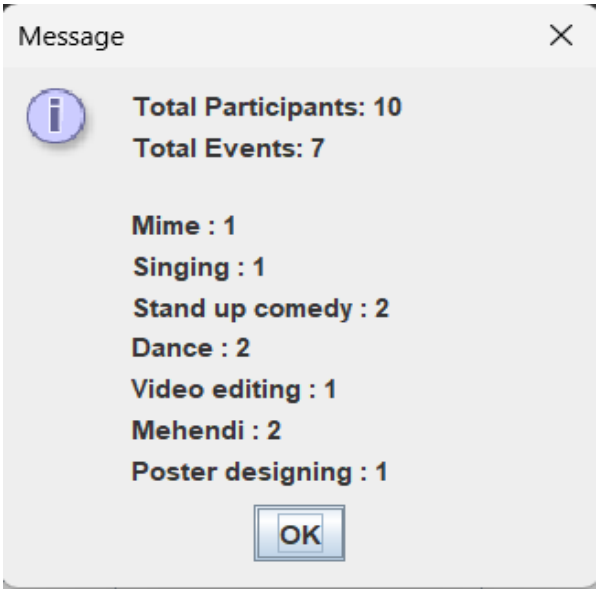


Figure 4: Dashboard / Summary Screen

Shows the participant statistics for all events.



7. DISCUSSION & CONCLUSION

7.1 Discussion

The Event Registration and Participant Management System was developed to automate the registration and management of participants in college events. The system allows users to register participants, update or delete records, search for participants, filter by event, and view overall statistics through a dashboard.

During the development process, **Core Java programming concepts** such as **loops, conditional statements, methods, ArrayLists, file handling, and object-oriented principles** were applied. The system successfully manages participant data, validates input, and maintains organized records.

The system was tested with different inputs to ensure that all functionalities—**registration, search, update, delete, filtering, and dashboard statistics**—perform correctly. The results demonstrate that the system reduces **manual effort**, improves **data management efficiency**, and helps organizers maintain accurate participant records.

7.2 Conclusion

The Event Registration and Participant Management System successfully demonstrates the practical application of **Core Java programming concepts** in developing a real-world application for managing college or organizational events.

The system automates **participant registration**, **data management**, and **event tracking**, reducing manual effort and improving record accuracy. It simplifies event operations by maintaining organized participant details, generating dashboards, and providing easy search and update functionalities.

This project helps students understand how **Java programming** can be used to develop practical software systems. The project can be further enhanced by adding features such as **database connectivity**, **advanced graphical user interfaces**, **online registration**, and **report generation**.

8. REFERENCES

The following resources were used during the development of this project:

Books

• Cay S. Horstmann – *Core Java Volume I – Fundamentals*

Websites

• https://www.oracle.com/java
• https://www.geeksforgeeks.org
• https://www.w3schools.com
• https://www.javatpoint.com